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Department	Mechanical Engineering

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Lab 01: Create a Simple Network

Date Performed	25-03-2026
Tool Used	Cisco Packet Tracer 9.0, Switch (2960), PCs, Straight-through cables
Status	Completed

Objective

Build a basic network topology by connecting end devices (PCs) to a switch and verifying connectivity using ICMP ping.

Network Topology

2 PCs connected to a Cisco 2960 switch via FastEthernet interfaces.

Steps Performed

- Opened Cisco Packet Tracer and created a new project.
- Added two PCs and one Cisco 2960 switch to the workspace.
- Connected PC0 to Switch (Fa0/1) and PC1 to Switch (Fa0/2) using straight-through cables.
- Configured IP addresses: PC0 → 192.168.1.1/24, PC1 → 192.168.1.2/24.
- Used the ping command from PC0 to verify connectivity to PC1.
- Observed successful ICMP echo request and reply.

Outcome

Successfully established basic LAN connectivity between two PCs via a switch. Ping responses confirmed proper IP configuration and Layer 2 switching.

Key Concepts Learned

OSI Layer 2 switching, IP addressing, ICMP, straight-through cables, LAN fundamentals.

Lab 02: Create a Simple Network 2

Date Performed	25-03-2026
Tool Used	Cisco Packet Tracer 9.0, Switch (2960), Multiple PCs, Cables
Status	Completed

Objective

Extend the basic network by adding more end devices and verifying inter-device communication across a switch.

Network Topology

Multiple PCs connected to a single Cisco 2960 switch forming a star topology LAN.

Steps Performed

- Added additional PCs to the existing simple network topology.
- Connected all PCs to the switch using straight-through copper cables.
- Assigned unique IP addresses in the 192.168.1.0/24 subnet to each PC.
- Verified connectivity between all PC pairs using the ping utility.
- Checked MAC address table on the switch to confirm device learning.
- Observed how broadcast traffic is handled by the switch.

Outcome

Successfully created an expanded LAN with multiple devices. All devices communicated successfully, demonstrating Layer 2 MAC learning and forwarding.

Key Concepts Learned

MAC address table, switch forwarding, broadcast domain, IP subnetting, star topology.

Lab 03: Connect Devices using Wireless Technologies

Date Performed	25-03-2026
Tool Used	Cisco Packet Tracer 9.0, Wireless Router (Linksys WRT300N), Laptops with wireless adapters
Status	Completed

Objective

Configure a wireless network using a home router/access point and connect wireless end devices to it.

Network Topology

Wireless router connected to a modem; laptops connected wirelessly to the access point.

Steps Performed

- Added a Linksys WRT300N wireless router to the workspace.
- Configured the SSID and WPA2 security key on the wireless router.
- Added laptops with wireless NIC modules (WPC300N).
- Connected laptops to the wireless network using the configured SSID and password.
- Verified DHCP-assigned IP addresses on each wireless device.
- Tested connectivity using ping between wireless devices and the router.

Outcome

Successfully configured a wireless LAN (WLAN). Laptops connected to the access point wirelessly and received DHCP addresses, with full connectivity verified.

Key Concepts Learned

WLAN, SSID, WPA2 security, DHCP, wireless NIC, access point, 802.11 standards.

Lab 04: Create Realistic Structured Cabling in the Physical Workspace

Date Performed	25-03-2026
Tool Used	Cisco Packet Tracer 9.0, Physical Workspace View, Patch panels, Switches, Rack units
Status	Completed

Objective

Simulate a real-world structured cabling environment using physical workspace view and mount devices in a server rack.

Network Topology

Server rack with mounted switches, patch panels, and structured cabling connecting devices in a physical building layout.

Steps Performed

- Switched to Physical Workspace view in Cisco Packet Tracer.
- Navigated into a building and placed network devices in the IDF/MDF room.
- Mounted a Cisco switch and patch panel into the server rack.
- Used structured cabling to connect wall ports to patch panel ports.
- Connected patch panel to the switch using patch cables.
- Verified device placement and cable management in the rack view.
- Tested logical connectivity between devices after physical setup.

Outcome

Successfully simulated enterprise-grade structured cabling in a physical workspace. Understood real-world cable management, rack mounting, and IDF/MDF concepts.

Key Concepts Learned

Structured cabling, patch panels, IDF/MDF, server racks, physical vs logical topology, TIA/EIA standards.

Lab 05: Edit Topologies

Date Performed	30-03-2026
Tool Used	Cisco Packet Tracer 9.0, Existing topology files, Various network devices
Status	Completed

Objective

Modify an existing network topology by adding, removing, and reconfiguring network devices and connections.

Network Topology

Pre-built topology modified by adding new devices, changing connections, and reconfiguring IP addresses.

Steps Performed

- Opened an existing Packet Tracer topology file.
- Identified devices to be added, removed, or modified.
- Added new switches and PCs to extend the network.
- Deleted unused or incorrect connections and replaced with proper cable types.
- Updated IP addressing scheme to accommodate new devices.
- Verified updated topology using ping and traceroute.
- Saved the modified topology as a new file.

Outcome

Successfully edited and updated network topology. Demonstrated the ability to modify live network designs and maintain connectivity after changes.

Key Concepts Learned

Topology modification, network scalability, cable types, IP reconfiguration, traceroute.

Lab 06: Explore Device Configuration Using the CLI (Console)

Date Performed	26-03-2026
Tool Used	Cisco Packet Tracer 9.0, Cisco Router (1841/2811), Cisco Switch (2960), Console cable, Terminal
Status	Completed

Objective

Access and configure a Cisco router/switch using the Command Line Interface (CLI) via console connection.

Network Topology

PC connected to a Cisco router via console (rollover) cable for out-of-band management.

Steps Performed

- Connected a PC to the router using a console (rollover) cable.
- Opened terminal emulation on the PC (9600 baud, 8N1).
- Accessed the router CLI and entered privileged EXEC mode using 'enable'.
- Entered global configuration mode using 'configure terminal'.
- Set hostname: hostname RouterA.
- Configured enable password and console password for security.
- Assigned IP address to FastEthernet interface: ip address 192.168.1.1 255.255.255.0.
- Brought up the interface using 'no shutdown'.
- Saved configuration using 'write memory' / 'copy run start'.
- Verified configuration using 'show running-config' and 'show interfaces'.

Outcome

Successfully configured a Cisco device via CLI through console access. Understood IOS command hierarchy, interface configuration, and configuration persistence.

Key Concepts Learned

Cisco IOS, CLI modes (User/Privileged/Global Config), console access, interface configuration, show commands, running vs startup config.

Lab 07: Manage and Configure your Network using a Network Controller

Date Performed	30-03-2026
Tool Used	Cisco Packet Tracer 9.0, Network Controller, Managed switches, PCs
Status	Completed

Objective

Use a network controller (SDN-based) to centrally manage and configure network devices.

Network Topology

Network controller connected to multiple switches; centralized management of network policies and device configurations.

Steps Performed

- Added a Network Controller device to the topology.
- Connected switches to the network controller for centralized management.
- Accessed the network controller GUI through a web browser on a PC.
- Discovered and added network devices to the controller dashboard.
- Applied network policies and VLAN configurations through the controller interface.
- Monitored device status and network health from the controller dashboard.
- Verified that configurations pushed from the controller were applied to devices.

Outcome

Successfully managed network devices using a centralized SDN controller. Understood the difference between traditional CLI management and controller-based network management.

Key Concepts Learned

SDN (Software Defined Networking), network controller, centralized management, network policies, VLAN configuration, northbound/southbound APIs.

Lab 08: Modify an Existing Script for an IoT Thing

Date Performed	07-04-2026
Tool Used	Cisco Packet Tracer 9.0, IoT devices (sensors/actuators), Script editor, Python/JavaScript
Status	Completed

Objective

Edit and customize an existing Python/JavaScript script to modify the behavior of an IoT device in Packet Tracer.

Network Topology

IoT devices connected to a home gateway; scripted automation controlling device behavior.

Steps Performed

- Opened an existing IoT topology with pre-configured devices and scripts.
- Identified the IoT device with an existing automation script.
- Opened the script editor within the IoT device properties.
- Analyzed the existing script logic and identified parameters to modify.
- Changed trigger conditions (e.g., temperature threshold, motion detection sensitivity).
- Modified output actions (e.g., turning on/off lights, fans, alarms).
- Saved the modified script and ran simulation to test changes.
- Observed the updated IoT device behavior matching modified script logic.

Outcome

Successfully modified IoT automation scripts. Understood event-driven programming in IoT context and how scripts control physical device behavior.

Key Concepts Learned

IoT scripting, event-driven programming, Python/JS in Packet Tracer, IoT sensors/actuators, automation logic, home gateway.

Lab 09: Create Your Own IoT Thing

Date Performed	07-04-2026
Tool Used	Cisco Packet Tracer 9.0, IoT Custom Thing editor, Python scripting, IoT gateway
Status	Completed

Objective

Design and build a custom IoT device from scratch by defining its components, behaviors, and automation scripts.

Network Topology

Custom IoT device integrated into a smart home network, connected via home gateway to control interface.

Steps Performed

- Opened the IoT Custom Thing editor in Cisco Packet Tracer.
- Defined the custom device name, type, and visual representation.
- Added input components (sensors: temperature, humidity, motion).
- Added output components (actuators: LED, buzzer, servo motor).
- Wrote a custom Python script to define device logic and automation rules.
- Connected the custom IoT device to the home gateway.
- Registered the device in the IoT server/controller.
- Tested the custom device by triggering sensor inputs and observing actuator responses.
- Verified device status on the IoT monitoring dashboard.

Outcome

Successfully created a fully functional custom IoT device with defined sensors, actuators, and automation logic. Demonstrated understanding of IoT device design and programming.

Key Concepts Learned

Custom IoT development, sensor/actuator design, Python scripting, IoT gateway, device registration, smart home automation.

Lab 10: Monitor Your Network Using a Network Controller

Date Performed	30-03-2026
Tool Used	Cisco Packet Tracer 9.0, Network Controller, Managed switches/routers, PCs
Status	Completed

Objective

Use a network controller's monitoring features to observe network health, traffic, and device status in real time.

Network Topology

Full network with controller monitoring multiple switches and routers, with PCs generating traffic.

Steps Performed

- Set up a network with multiple switches, routers, and end devices.
- Connected all devices to the network controller for centralized monitoring.
- Accessed the network controller monitoring dashboard via web browser.
- Observed real-time device status indicators (up/down, warnings).
- Generated network traffic using ping and file transfers between PCs.
- Monitored bandwidth utilization and interface statistics on the dashboard.
- Identified any network anomalies or connectivity issues via the controller.
- Used controller tools to drill down into specific device configurations and logs.

Outcome

Successfully monitored a multi-device network using the controller dashboard. Gained hands-on experience with centralized network visibility and real-time status monitoring.

Key Concepts Learned

Network monitoring, SDN controller dashboard, real-time analytics, interface statistics, network health, fault detection.

Summary

This report documents the completion of 10 Cisco Packet Tracer hands-on labs as part of the Cisco NetAcad curriculum. The labs covered a wide range of networking topics from basic LAN setup to advanced IoT scripting and SDN-based network management.

Skills Demonstrated

- Network design and topology creation using Cisco Packet Tracer 9.0
- IP addressing, subnetting, and DHCP configuration
- Cisco IOS CLI configuration and device management
- Wireless network setup with WPA2 security
- Structured cabling and physical workspace management
- SDN-based network monitoring and policy management
- IoT device creation, scripting, and automation

Tools and Technologies Used

- Cisco Packet Tracer 9.0.0
- Cisco IOS CLI (routers and switches)
- Python / JavaScript (IoT scripting)
- SDN Network Controller
- Wireless technologies (802.11, WPA2)

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